The Most Efficient Protocol for PAKE:
What Exact Stuff Do You Need to Hash at the End?

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1 Introduction

2 Security Analysis of the "Hash Everything" Version

2.1 The Simulator

2.1.1 Simulation Strategy

A large portion of the simulator is routine; below we explain the idea behind the non-trivial parts.

Simulating honest P-to-P' message. When P's session begins, $\mathcal{S}$ must simulate its message (s, T) to P'. This can be easily done: just sample (s, T) at random. However, every time $\mathcal{A}$ queries $H_2(\mathsf{pw}^*, T)$ (let the result be t^*), $\mathcal{S}$ needs to generate a KEM key pair (pk^*, sk^*) for later use (see the "Extracting password guess from malicious P'" paragraph below) and make sure that (pk^*, sk^*) is consistent with the RO queries. This can be done as follows: $\mathcal{S}$ computes

$$r^* := s \oplus t^*, \quad R^* := T/pk^*,$$

and programs $H_1(\mathsf{pw}^*, r^*) := R^*$. This also allows $\mathcal{S}$ to store the association between pw^* and (pk^*, sk^*). (Note that (pk^*, sk^*) needs to be freshly generated every time $\mathcal{A}$ queries $H_2(\star, T)$; otherwise R^* never changes and $\mathcal{A}$ can easily find collisions in H_1.)

Extracting password guess from malicious P. When the adversary $\mathcal{A}$ sends a message (s^*, T^*) to P', if it is different from the honest message (s, T), the simulator $\mathcal{S}$ must extract a password guess (if any) and send a corresponding $\mathsf{TestPwd}$ command to $\mathcal{F}_{\text{PAKE}}$. There are two types of attack, which need to be addressed separately:

1. *"Normal" online attack:* $\mathcal{A}$ executes P's algorithm on a password guess pw^*. That is, $\mathcal{A}$ samples public key pk^* and randomness r^*, computes

$$R^* := H_1(\mathsf{pw}^*, r^*), \qquad T^* := pk^* \cdot R^*,$$
$$t^* := H_2(\mathsf{pw}^*, T^*), \qquad s^* := t^* \oplus r^*,$$

   and sends (s^*, T^*) to P'.
   It takes a while to realize how to extract pw^* from (s^*, T^*). First, $\mathcal{S}$ should scan all $H_2(\star, T^*)$ queries, which yields multiple pw^* values (and multiple t^* values); $\mathcal{S}$ must decide which one is the "actual" password guess. This can be done via the following. For each such (pw^*, t^*) pair, $\mathcal{S}$ computes the corresponding $r^* := s^* \oplus t^*$, and checks if (and when) $H_1(\mathsf{pw}^*, r^*)$ was queried. *With overwhelming probability, there is at most one such H_1 query made before the $H_2(\mathsf{pw}^*, T^*)$ query*, from which $\mathcal{S}$ can extract the "actual" password guess pw^*. (This special $H_1(\mathsf{pw}^*, r^*)$ query is called the "anchor query" in [MRR20, JRX25].)

2. *Related key attack:* This attack works only after $\mathcal{A}$ receive an honest (s, T) pair from P. $\mathcal{A}$ skips sampling pk^* or r^*; instead, $\mathcal{A}$ picks any $\Delta \in \mathbb{G}$, computes $T^* := T \cdot \Delta$ and

$$t^* := H_2(\mathsf{pw}^*, T), \qquad (t^*)' := H_2(\mathsf{pw}^*, T^*),$$
$$\delta := t^* \oplus (t')^*, \qquad s^* := s \oplus \delta,$$

   and sends (s^*, T^*) to P'.
   Let us first explain what $\mathcal{A}$ is trying to achieve here. The same randomness r yields two valid messages, (s, T) and (s^*, T^*), such that

$$s \oplus s^* = H_2(\mathsf{pw}, T) \oplus H_2(\mathsf{pw}, T^*),$$

   which allows $\mathcal{A}$ to pick T^* and "reverse engineer" the corresponding s^* if it guesses pw correctly. (s^*, T^*) implicitly encrypts $pk^* = T^*/H_1(\mathsf{pw}, r)$, which $\mathcal{A}$ can compute as $pk \cdot (T^*/T)$.
   To extract pw^* from (s^*, T^*), $\mathcal{S}$ should scan all $H_2(\star, T)$ and $H_2(\star, T^*)$ queries (which yield multiple candidate pw^* values) and check which ones satisfy

$$H_2(\mathsf{pw}^*, T) \oplus H_2(\mathsf{pw}^*, T^*) = s \oplus s^*.$$

   With overwhelming probability, at most one pw^* will satisfy this equation.

Extracting password guess from malicious P'. When the adversary $\mathcal{A}$ sends a message (c^*, τ^*) to P, if $\mathcal{A}$ is not passive (i.e., it is not the case that $(s^*, T^*) = (s, T)$ *and* $(c^*, \tau^*) = (c, \tau)$), the simulator $\mathcal{S}$ must extract a password guess (if any) and send a corresponding $\mathsf{TestPwd}$ command to $\mathcal{F}_{\text{PAKE}}$.

$\mathcal{S}$ should find the H_{tag} query whose result is τ^*, which includes $(\mathsf{pw}^*, pk^*, K^*)$. However, pw^* constitutes a password guess only if it is consistent with pk^* and K^*, i.e., $K^* = \mathsf{Decap}(sk^*, c^*)$ for sk^* corresponding to pk^*. $\mathcal{S}$ can check this

condition because it has stored the correspondence between pw^* and (pk^*, sk^*) (see the "Simulating honest P-to-P' message" paragraph above), and can use sk^* to check if the equation holds. (This extraction strategy does not work anymore if pw^* is not included in H_{tag}. In later sections we will explain how to simulate the "not hashing pw" version of the protocol.)

References

[JRX25] Jake Januzelli, Lawrence Roy, and Jiayu Xu. Under what conditions is encrypted key exchange actually secure? In *EUROCRYPT 2025*, pages 451–481, 2025.

[MRR20] Ian McQuoid, Mike Rosulek, and Lawrence Roy. Minimal symmetric PAKE and 1-out-of-N OT from programmable-once public functions. In *CCS 2020*, pages 425–442, 2020.